Lab – IoT data

Smart Indoor Climate System

Background
Sensors produce autonomous data which can be collected continuously or periodically. Sensors work like a mini-computer with their own set of operations. Working with sensor data helps us automate actions, for example turning heating or cooling devices ON or OFF automatically based on the observed and predicted conditions.

Purpose
The purpose of this lab is to learn how to analyse and forecast sensor data. Sensors produce data which can be collected over a period of time. This historical data can be used to explore patterns and to build models that predict future values. You will use these predictions to design a simple control system for indoor climate.

Dataset
Use the following open dataset:
Appliances Energy Prediction – https://www.kaggle.com/datasets/loveall/appliances-energy-prediction

Important columns:
- Appliances: Energy consumption of appliances (Wh).
- T1: Indoor temperature (kitchen area).
- T_out: Outdoor temperature.
- RH_1, RH_out: Indoor and outdoor humidity.
- date: Timestamp of the measurement.

Lab Overview
You will complete three main parts:
1. Task 1 – Data understanding and preprocessing.
2. Task 2 – Forecasting indoor temperature or energy use.
3. Task 3 – Designing a simple forecast-based control system.

Use Python (pandas, matplotlib, and any basic machine learning library such as scikit-learn or statsmodels).

Task 1 – Data Understanding & Preprocessing

- Load the dataset from the CSV file.
- Parse the 'date' column as a datetime type and set it as the index.

- Remove or handle missing values.
- Create simple time-based features, for example:
 • hour of the day
 • day of the week
 • weekend (yes/no)
- Create lag features that use previous values, for example:
 • T1_lag1: previous value of T1
 • T_out_lag1: previous value of T_out
- Perform basic exploratory analysis:
 • Plot indoor vs outdoor temperature over time (T1 and T_out).
 • Plot the distribution of appliance energy use (Appliances).
 • Show a simple correlation matrix between T1, T_out and Appliances.

Task 2 – Forecasting Module

Build a simple model to forecast either indoor temperature (T1).

- Choose one target variable to predict: T1 or other.
- Use a time-based train/test split (for example, first 80% of time steps for training, last 20% for testing).

Select input features such as:
 • T_out, hour, day, weekend
 • lagged values like T1_lag1, T_out_lag1

Train at least one simple forecasting model, for example:
 • Linear Regression
 • Random Forest Regression

Evaluate the model on the test set using:
 • Mean Absolute Error (MAE)
 • Root Mean Squared Error (RMSE)
- Plot the true values and predicted values in the same figure for a sample period.

Task 3 – Forecast-Based Control System

Use your forecast to control a simple heating system.

Assume:
- T1 is the indoor temperature.
- We want to keep T1 roughly between 20°C and 23°C.
- A heating device can be either ON (1) or OFF (0).

Steps:
- Use the predictions from your model in Task 2 (for example, forecasted T1).

For each time step in the test period, apply simple rules such as:

- If forecasted T1 < 20°C → set heating to ON.
- If forecasted T1 > 23°C → set heating to OFF.
- Otherwise, keep the previous heating state.

- Save the heating state (0 or 1) for each time step.
- Plot both the actual indoor temperature (T1) and the heating state over time. Highlight in the plot when the heating state changes.

Hand-in: Deadline 10/Dec 23:59

Record a video presentation explaining the three tasks. All members in the group must be involved in the presentation. The maximum time for the recording is 10 minutes.

- A short description of what you observe in the plots and a cleaned dataset ready for forecasting. Explanation of which model you used, the error values (MAE and RMSE), and at plot showing true vs predicted values. Finally, explain how your controller behaves, the two metrics above, and plots showing temperature and heating state.
- Include a short report with all the figures mentioned in the three tasks.

Useful links:

https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html

https://www.geeksforgeeks.org/machine-learning/random-forest-regression-in-python/

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

https://www.geeksforgeeks.org/machine-learning/ml-linear-regression/